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MATP32

Exercises 2

1. Prove the Minkowski Integral Inequality:

Let $(X, \mathcal{A}, \mu)$, $(Y, \mathcal{B}, \nu)$ be σ -finite measure spaces, $1 \leq p < \infty$. Then for any $\mathcal{A} \times \mathcal{B}$ -measurable function $f : X \times Y \rightarrow \mathbb{C}$ with the property that $f(\cdot, y)$ is μ -integrable for ν -almost $y \in Y$, we have

$$\left(\int_Y \left| \int_X f(x, y) d\mu(x) \right|^p d\nu(y) \right)^{1/p} \leq \int_X \left(\int_Y |f(x, y)|^p d\nu(y) \right)^{1/p} d\mu(x).$$

2. Let $g : [0, \infty) \rightarrow [0, \infty)$ such that g is decreasing and has only finitely many discontinuities.

a) Define

$$k : \mathbb{R}^d \rightarrow [0, \infty), \quad k(x) = g(|x|) \text{ for } x \in \mathbb{R}^d.$$

Suppose that $k \in L^1(\mathbb{R}^d)$. Show that for every $f \in L^1_{loc}(\mathbb{R}^d)$,

$$\sup_{t>0} (|f| * k_t)(x) \leq \|k\|_{L^1} M_c f(x) \text{ for a.e. } x \in \mathbb{R}^d.$$

- b) Now suppose that $k \in L^1(\mathbb{R}^d)$ with $\int f(x) dx = 1$ and $|k(x)| \leq g(|x|)$ for all $x \in \mathbb{R}^d$. Show that for any $p \in [1, \infty)$ and any $f \in L^p(\mathbb{R}^d)$,

$$\lim_{t \rightarrow 0^+} (k_t * f)(x) = f(x) \text{ for a.e. } x \in \mathbb{R}^d.$$

3. Let $p \in [1, \infty)$, $f \in L^p(\mathbb{R}^d)$. We define

$$u : \mathbb{R}^d \times (0, \infty) \rightarrow \mathbb{C}, \quad u(x, t) = P_t * f(x),$$

where P is the Poisson kernel given by

$$P(x) = \frac{\Gamma((d+1)/2)}{\pi^{(d+1)/2}} \frac{1}{(1 + |x|^2)^{(d+1)/2}} \quad (x \in \mathbb{R}^d).$$

Show that u solves the Dirichlet problem on the half-space $\mathbb{R}^{d+1}_+ = \mathbb{R}^d \times (0, \infty)$ with boundary values f in the following sense:

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a) Show that u is harmonic on $\mathbb{R}_+^{d+1}$, i.e.

$$\partial_t u(x, t) + \sum_{i=1}^d \partial_i^2 u(x, t) = 0 \text{ for } (x, t) \in \mathbb{R}_+^{d+1}.$$

b) Show that

$$\lim_{t \rightarrow 0^+} u(x, t) = f(x) \text{ for a.e. } x \in \mathbb{R}^d.$$

4. a) Show that the functional $p.v.(1/x)$ given by

$$\langle p.v.(1/x), \phi \rangle = \lim_{\varepsilon \rightarrow 0^+} \int_{|x| > \varepsilon} \frac{\phi(x)}{x} dx \quad (\phi \in \mathcal{S}(\mathbb{R}^d))$$

is well-defined (i.e. the limit above exists), and a bounded linear functional on $\mathcal{S}(\mathbb{R}^d)$, hence a tempered distribution.

b) Show that

$$\langle \widehat{p.v.(1/x)}, \phi \rangle = \langle -i\pi \operatorname{sgn}, \hat{\phi} \rangle,$$

where sgn is given by

$$\operatorname{sgn}(\xi) = \begin{cases} 1 & \text{for } \xi > 0 \\ 0 & \text{for } \xi = 0 \\ -1 & \text{for } \xi < 0. \end{cases}$$

The questions will be discussed in the seminar on Thursday, September 17.